

Curriculum Vitae of Dr. Niraj Ramesh Dayama

Summary

My curriculum vitae presents a candidate with a blend of proven research skills, software delivery experience and IT consultancy expertise. I am a post-doctoral researcher with an extensive experience in IT and software fields. My software consultancy competence covers IT project management and system design across the areas of mathematical programming, combinatorial optimization and computational engineering. I have handled industry-sponsored research projects in industry and academia; I also managed classical turnkey software delivery projects across multiple industry segments.

Personal Details

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Academic Qualification

Qualification	Affiliation	Performance	Time period
<i>Doctor of Philosophy</i> in Operations Research	Monash Uni Australia / IIT Bombay IITB-Monash Research Academy	CGPA → 8.3 / 10	Jan 2010 to Aug 2014
<i>Master of Technology</i> in Computational Engg.	Indian Institute of Technology Madras Chennai, India	CGPA → 9.3/10	Jul 2005 to Jul 2007
<i>Bachelor of Engineering</i> in Mechanical Engg.	Pune University, Pune, India	Percentage → 70%	Jul 1997 to Jun 2001

Highlights of Current Employment and Professional Role

University of Jyväskylä	I have been working as a post-doctoral researcher since 2023. This job qualifies as a stage 3 researcher according to Academic of Finland career model. My current research involves industry-sponsored projects to explore quantum computing.
Arcada University of Applied Sciences	I teach graduate level classes in Arcada UAS since 2023. This role involves teaching 10 subjects annually within IT and computer science streams.
Tata Consultancy Services	I have been engaged with consultancy and IT project delivery management for TCS Finland since 2014. In this role, I have been a software delivery manager, architect and consultant working for multiple Finnish companies: Nordea, Wäertsilä, etc.

Previous Professional and Academic Background

Time	Organization Affiliation and Role	Responsibility and Major Outcomes
2017 to 2022	Aalto University Post-doctoral researcher	- Teacher for graduate level classes on applied mathematical and optimization techniques - Handled 4 externally-sponsored research projects leading to 10+ publications.
2014 to Ongoing	Tata Consultancy Services, Finland Corporate Trainer and IT Project Delivery manager	- Corporate Trainer of software architecture - Certified Scum master. IT Project delivery manager - Executed turnkey software IT projects for prominent Finnish customers (Manufacturing and Banking)
2010 to 2013	IIT Bombay & Monash University Ph.D. Student Mathematical programming	- Published 12 research papers at relevant forums. - Participated in inter-disciplinary research for grant proposals and industry-sponsored research projects
2007 to 2010	Tata Consultancy Services, India System Analyst and Designer Trainer for Software System Design	- Corporate Trainer of three subjects: Java, Data Structures/Algorithms and Corporate soft skills - Technical lead and consultant for Design and development of IT systems at Siemens PLM client site.
2007 to 2011	Pune University Adjunct faculty member and external examiner	- Undergraduate and graduate-level courses including <i>Database management</i> and <i>Algorithm Design</i> - Supervisor and designated external examiner for final-year projects executed by students
2005 to 2007	Indian Inst. Of Tech. Madras, India Master's Student (M.Tech.) Computational Engineering	- Developed a novel algorithm for fluid flow simulation. - Master's Degree with CGPA of 9.3/10 - Teaching assistant for two undergraduate courses
2001 to 2005	IT Consultancy Organizations, India Software Developer Software Development and Testing	- Certified Java developer working on software projects - Learnt IT design, DBMS and process automation - Executed projects in logistics & automotive domain
1997 to 2001	Pune University, India Bachelor's Undergraduate Student	- Best outgoing student award earned due to combination of academic and extra-curricular skills - Internship with Mercedes-Benz (Audit and process)

* At several stages, I have worked in research or academics, while also doing a software consultancy job.

Professional Skillset

Primary Competencies	Applied mathematical programming, machine learning, combinatorial optimization Software development, Project management and end-to-end project execution Proposal preparation for industrial projects and research grants
Preferred Tools	IBM Cplex, Gurobi, Arena Simulation, IBM Rational, Oracle Designer, Eclipse
Software Languages	Java (including J2EE platform) and Python, R (Analytics)
Methods	Algorithms design, UI/UX development, mixed integer programming, Software architecture and design, professional software development process (Agile/Scrum)

Major Teaching Experience

Organization Role and time period	Details of Responsibility Handled
Aalto University Post-Doctoral Researcher 2017-2022	- I have taught Computational User Interface Design course over 4 academic years. This was a graduate-level course. - This course involved applied optimization techniques including integer programming and combinatorial optimization. - I handled all aspects of my course including the content creation, actual classroom presentation, assignments and evaluation.
Tata Consultancy Services Corporate trainer 2014-2016	- I have taught various courses in software architecture, software design, Java programming language and Data structures & Algorithms. - Typical class strength was 50-75 attendees. I specifically conducted 3-4 hour intensive sessions on alternate days. - I also conducted specific lectures on non-technical topics such as corporate soft-skills and on very niche topics such as PLM. - Attendees were corporate employees of companies including Tata Consultancy Services, CG Pauwels Belgium and of Siemens PLM. - I have taught in various geographies including India, Germany, Belgium and Finland.
Pune University, India Adjunct Faculty 2008-2011	- I have taught two courses at Masters (graduate) level over 3 academic years: Database management systems and Design of Algorithms. - Every course required 32-40 contact hours of classroom teaching followed by assignments and final evaluation.

Academic Supervision Experience

Venue	Role	Responsibility Handled
Aalto University	Supervisor / Thesis advisor	5 Masters' students
		3 Bachelor's students
		9 summer interns
Monash University	Thesis supervisor	3 Bachelor's students
Pune University	Thesis advisor / Guide External examiner	Guided 5 Masters' students
		External examiner for 16
		Bachelor's students

External Research Funding Achievements

External funding source	Business Finland	Funding amount	€ 950 000
Affiliation organization (fund recipient)	Aalto University	Principal Investigator	Prof Antti Oulasvirta
Project Title	Design.AI	Application Period	April 2021
My specific role	▪ Primary driver for the governing research concept ▪ Sole presenter to the funding source for the fund application ▪ Key contributor (along with PI) for composing proposal content		
External funding source	Huawei Tech.	Funding amount	€ 350 000
Affiliation organization (fund recipient)	Aalto University	Principal Investigator	Prof Antti Oulasvirta
Project Title	Style Transfer	Application Period	Oct 2019
My specific role	▪ Proactively introduce the project concept to the industry partner ▪ Key contributor (along with PI) for composing proposal content ▪ Project manager from initial estimation upto final execution		
External funding source	Australian Research Council (ARC Grant)	Funding amount	Au\$ 900 000 Equal to €600 000
Affiliation organization (fund recipient)	Monash University in Australia	Principal Investigator	Dr Andreas Ernst and Prof Mohan K
Project Title	Workforce/Logistic scheduling for modern workspace	Application Period	Oct 2015
My specific role	▪ I made the high-level solution design required for the proposal ▪ I also owned the detailed effort estimation and wrote the literature survey of the proposal		

In addition to the approved/achieved funds listed above, I have also participated in research proposals and grant applications for other initiatives. This includes early exploration of incubation and business-launch discussion with NewCo Helsinki, funding proposal to Business Finland, etc.

Awards and Honors

1. Best thesis talk: Certificate and award from IIT Bombay (2011) and Monash University (2013).
2. Best student paper: Certificate from Australian society of operations research (2013)
3. Special initiative award for teaching at Siemens PLM during 2008/09
4. Business Plan Competition: Certificate and award from IIT Bombay in B-PlanZ 2006
5. Best Outgoing Student: Award from under-graduate degree at VIT Pune, India in 2001/01.